

Theorem (Cauchy–Schwarz). For real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

Equality holds iff there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i (including the trivial case $b_1 = \dots = b_n = 0$ or $a_1 = \dots = a_n = 0$).

Proof. **Degenerate case** $\sum_{i=1}^n b_i^2 = 0$. If $\sum_{i=1}^n b_i^2 = 0$ then each $b_i = 0$. Consequently

$$\sum_{i=1}^n a_i b_i = 0, \quad \left(\sum_{i=1}^n a_i b_i \right)^2 = 0, \quad \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) = 0,$$

so the inequality holds with equality. From now on assume $\sum_{i=1}^n b_i^2 > 0$.

Construction of a non-negative quadratic. For an arbitrary real parameter t define

$$Q(t) := \sum_{i=1}^n (a_i - t b_i)^2.$$

Each summand is a square, hence $Q(t) \geq 0$ for all $t \in \mathbb{R}$.

Expanding and collecting powers of t gives

$$\begin{aligned} Q(t) &= \sum_{i=1}^n (a_i^2 - 2t a_i b_i + t^2 b_i^2) \\ &= \underbrace{\sum_{i=1}^n b_i^2}_{A} t^2 + \underbrace{(-2 \sum_{i=1}^n a_i b_i)}_{B} t + \underbrace{\sum_{i=1}^n a_i^2}_{C}. \end{aligned}$$

Thus $Q(t) = At^2 + Bt + C$ with

$$A = \sum_{i=1}^n b_i^2 > 0, \quad B = -2 \sum_{i=1}^n a_i b_i, \quad C = \sum_{i=1}^n a_i^2.$$

Discriminant condition for a non-negative quadratic. *Lemma.* If a quadratic $f(t) = At^2 + Bt + C$ satisfies $f(t) \geq 0$ for all $t \in \mathbb{R}$ and $A > 0$, then its discriminant is non-positive: $B^2 - 4AC \leq 0$.

Proof of Lemma. If $B^2 - 4AC > 0$ the polynomial would have two distinct real roots and would be negative between them, contradicting $f(t) \geq 0$. If $B^2 - 4AC = 0$ the polynomial has a double root and is non-negative; if $B^2 - 4AC < 0$ it has no real root and is strictly positive.

Applying the lemma to $Q(t)$ yields

$$B^2 - 4AC \leq 0.$$

Substituting the expressions for A, B, C we obtain

$$\left(-2 \sum_{i=1}^n a_i b_i\right)^2 - 4 \left(\sum_{i=1}^n b_i^2\right) \left(\sum_{i=1}^n a_i^2\right) \leq 0.$$

Algebraic simplification. Using the elementary identity

$$\left(-2 \sum a_i b_i\right)^2 - 4 \left(\sum b_i^2\right) \left(\sum a_i^2\right) = 4 \left[\left(\sum a_i b_i\right)^2 - \left(\sum a_i^2\right) \left(\sum b_i^2\right) \right],$$

the previous inequality becomes

$$4 \left[\left(\sum a_i b_i\right)^2 - \left(\sum a_i^2\right) \left(\sum b_i^2\right) \right] \leq 0.$$

Dividing by the positive constant 4 gives the Cauchy–Schwarz inequality

$$\boxed{\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right)}.$$

Equality condition. Equality in the discriminant inequality occurs exactly when $B^2 - 4AC = 0$, i.e. when the quadratic $Q(t)$ has a double root. Hence there exists $t_0 \in \mathbb{R}$ such that $Q(t_0) = 0$. Since

$$Q(t_0) = \sum_{i=1}^n (a_i - t_0 b_i)^2 = 0,$$

each summand must be zero, giving

$$a_i = t_0 b_i \quad \text{for all } i = 1, \dots, n.$$

Conversely, if a constant λ satisfies $a_i = \lambda b_i$ for all i , then

$$\left(\sum a_i b_i\right)^2 = \left(\lambda \sum b_i^2\right)^2 = \lambda^2 \left(\sum b_i^2\right)^2 = \left(\sum a_i^2\right) \left(\sum b_i^2\right),$$

so equality holds. The case $\lambda = 0$ corresponds to the degenerate situation where one of the vectors is the zero vector.

Thus equality holds iff there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every index i (including the trivial zero-vector case). $\square$